

PORT ALBERNI, BC, Canada

WMO: 714750

Lat: 49.320N

Lon: 124.930W

Elev: 76

StdP: 100.41

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-6.8	-4.7	-9.7	1.7	-3.9	-7.0	2.1	-2.8	6.4	3.0	6.1	1.0	1.2	320	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	32.0	18.5	29.6	17.9	27.1	17.2	19.6	29.3	18.6	27.5	17.7	25.5	2.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	11.6	21.1	15.4	11.1	20.3	14.7	10.6	19.5	55.9	29.3	52.7	27.6	50.0	25.7	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
5.7	4.6	3.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-10.3	35.3	3.3	1.9	-12.6	36.8	-14.5	37.9	-16.3	39.0	-18.7	40.4
			-10.4	20.9	3.1	1.1	-12.7	21.8	-14.5	22.4	-16.3	23.0	-18.5	23.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	2.2	3.1	5.3	8.3	12.4	15.1	18.4	18.4	15.1	9.6	4.9	1.9		
	DBStd	6.59	3.10	2.64	2.49	2.73	3.18	2.92	2.81	2.64	2.69	2.68	3.36	2.99		
	HDD10.0	1108	242	193	146	66	12	1	0	0	1	39	157	252		
	HDD18.3	3273	501	426	403	301	184	105	34	32	102	271	404	509		
	CDD10.0	959	0	0	1	14	87	153	259	261	153	27	2	1		
	CDD18.3	82	0	0	0	0	1	8	34	34	4	0	0	0		
	CDH23.3	1603	0	0	0	9	82	198	597	573	143	1	0	1		
	CDH26.7	598	0	0	0	1	17	72	236	232	40	0	0	0		
(14) Wind	WSAvg	1.3	1.2	1.0	1.1	1.3	1.4	1.5	1.6	1.4	1.1	1.1	1.2	1.2		
(15) Precipitation	PrecAvg	1850	304	174	187	107	52	43	26	48	63	200	301	308		
	PrecMax	2295	574	479	366	234	149	105	66	214	225	443	660	480		
	PrecMin	1203	154	15	37	12	4	15	5	5	2	0	98	141		
	PrecStd	296	109	113	82	60	36	23	17	42	56	112	149	98		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	12.0	18.4	24.5	28.8	32.7	35.2	34.4	30.9	21.2	14.1	11.6		
		MCWB	10.1	9.0	10.9	14.1	15.8	18.2	20.2	18.9	18.0	14.1	12.7	10.6		
	2%	DB	9.5	10.1	13.7	19.8	25.6	28.6	32.0	31.9	27.2	17.9	12.3	8.7		
		MCWB	8.6	7.7	9.0	11.5	15.0	17.1	18.9	18.6	17.0	13.0	11.3	7.9		
	5%	DB	7.7	8.6	11.5	16.5	22.8	25.1	29.3	29.3	24.4	15.7	11.0	7.0		
		MCWB	7.1	6.9	8.1	10.2	14.1	16.2	18.2	18.0	16.1	12.6	10.1	6.5		
	10%	DB	6.1	7.1	9.9	14.1	19.7	22.1	26.7	26.5	21.7	14.0	9.8	5.6		
		MCWB	5.7	5.8	7.5	9.2	13.0	14.9	17.4	17.3	15.3	12.0	9.1	5.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	9.9	11.7	14.7	16.7	19.2	21.5	20.5	19.3	16.1	13.2	10.2		
		MCDB	11.0	11.0	15.3	21.9	25.7	29.9	33.4	31.1	28.1	18.5	13.8	10.8		
	2%	WB	8.4	8.3	10.0	12.5	15.5	17.8	20.0	19.3	17.9	14.2	11.5	7.6		
		MCDB	9.1	9.4	12.7	17.8	23.6	27.1	29.7	29.3	24.6	16.2	12.1	8.0		
	5%	WB	6.7	7.0	8.7	11.1	14.6	16.5	18.8	18.4	16.9	13.1	10.4	6.2		
		MCDB	7.2	8.1	10.6	14.7	21.5	24.0	27.6	27.3	22.4	14.9	10.9	6.6		
	10%	WB	5.5	5.9	7.7	9.8	13.4	15.3	17.7	17.6	15.9	12.1	9.2	5.1		
		MCDB	5.8	6.7	9.4	12.9	18.9	21.3	25.6	25.3	20.7	13.8	9.7	5.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	7.1	8.5	10.5	12.0	11.6	13.9	14.1	12.6	8.5	5.4	4.0		
		MCDBR	5.6	9.4	12.2	16.3	18.8	18.9	20.3	20.7	18.9	11.9	6.6	5.9		
		MCWBR	5.0	6.6	7.6	9.0	8.9	8.6	8.2	8.3	9.4	7.6	5.4	5.0		
	5% WB	MCDBR	5.5	7.8	10.1	13.3	16.2	16.7	18.1	18.1	15.6	10.1	6.0	5.3		
		MCWBR	4.8	6.0	6.9	7.8	7.9	7.7	7.8	7.7	8.1	6.9	5.1	4.7		
(40) Clear-Sky Solar Irradiance	taub		0.320	0.315	0.334	0.349	0.373	0.363	0.356	0.354	0.341	0.344	0.324	0.318		
	taud		2.459	2.539	2.497	2.450	2.393	2.444	2.504	2.517	2.545	2.494	2.474	2.414		
	Ebn at Noon		735	844	882	900	886	896	898	885	864	791	722	677		
	Edh at Noon		63	72	89	103	113	109	101	95	84	75	61	58		
(44) All-Sky Solar Radiation	RadAvg		0.84	1.74	2.60	4.03	5.20	5.63	6.12	5.22	3.72	2.02	1.01	0.66		
	RadStd		0.12	0.27	0.34	0.39	0.53	0.60	0.58	0.43	0.32	0.31	0.18	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+103	N/A
(47)	Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page